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North China block underwent simultaneous true polar wander and tectonic convergence in late Jurassic: New paleomagnetic constraints

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ABSTRACT

Jurassic witnessed two synchronous events on the Earth, named true polar wander (TPW) and closure of the Mongol–Okhotsk ocean (MOO) between Siberia and eastern Asian blocks. The paleogeographic change of the north China block (NCB) should thus include both TPW and plate tectonic components in Jurassic. We performed a new combined paleomagnetic and U–Pb geochronological study for Middle and Late Jurassic volcanic rocks from the northern NCB and obtained three new well-dated paleopoles: 82.7°N, 282.6°E, $A_{95} = 3.2°$ (ca. 160 Ma); 84.1°N, 228.9°E, $A_{95} = 6.6°$ (ca. 165 Ma); and 89.3°N, 216.8°E, $dp/dm = 5.4°/7.1°$ (ca. 170 Ma). A positive untilting test indicates a pre-folding origin interpretation for the ca. 160 Ma pole. Quality of these poles is enforced by three positive conglomerate tests, the presence of magnetic reversals and secular variation analyses. Comparison of the apparent polar wander paths of Eurasia and the NCB suggests that the NCB accreted to Eurasia around 140 Ma. The NCB and its then connected blocks did not undergo any significant southward paleogeographic shift during Jurassic. In context of TPW, we estimate a net northward plate moving velocity at 0.50–0.75°/myr for the NCB in Jurassic. The long-term accumulation of MOO subducted slabs at the high paleolatitudes may have critically contributed to, or perhaps led to, the Jurassic TPW.

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1. Introduction

True polar wander (TPW) is a migration of the entire mantle-crust system related to the rotation axis of Earth (Evans, 2003). A Jurassic TPW was revealed by a large body of paleomagnetic data from continental (e.g. Fu et al., 2020; Kent and Irving, 2010; Torsvik et al., 2012) and oceanic plates (Fu and Kent, 2018). The north China block (NCB) and its then connected blocks was separated from Siberia by the Mongol–Okhotsk ocean (MOO) in Jurassic (Ren et al., 2018, 2016; Van der Voo et al., 2015), which left an enigma to the paleogeographic change of the eastern Asian blocks during the Jurassic TPW. Some recent studies suggest that the NCB had already accreted onto Siberia and experienced a large southward latitudinal displacement as predicted by the "monster shift" TPW hypothesis during 174–157 Ma (Yi et al., 2019; Yi and Meert, 2020). This new model, mainly based on a new paleomagnetic pole obtained from the Nandaling Formation (Fm) exposed in the northern NCB, however, is profoundly in contrast to most previously published models (e.g. Huang et al., 2018; Ren et al., 2018, 2016; Van der Voo et al., 2015) and needs further test.

In this study, we report three robust poles obtained from the Middle and Late Jurassic lavas from the Haifanggou and Lanqi/Tiaojishan formations in the northern NCB. The new results suggest that there was no significant paleogeographic shift for the NCB in Jurassic. Integrating the updated paleomagnetic database with geological evidence, we propose a new kinematic model for the NCB and the convergence of MOO in the context of Jurassic TPW.

2. Geological setting and sampling

The NCB occupies a central portion of East Asia (Fig. 1a). It amalgamated with Amuria (AMU) block to the north in the late Paleozoic (Zhang et al., 2014; Ren et al., 2020), and with the south China block (SCB) during Late Triassic–Early Jurassic (Huang et al., 2018; Ren et al., 2016; Van der Voo et al., 2015). Our new paleomagnetic and geochronological investigation was carried out in the Beipiao region in western Liaoning Province, northern NCB (Fig. 1b), where the Jurassic to Early Cretaceous volcanic and sedimentary successions variously overlie Archean to Triassic rocks (Figs. 1c and 2; BGMRLP, 1989). The sampled Middle–Upper Juras-

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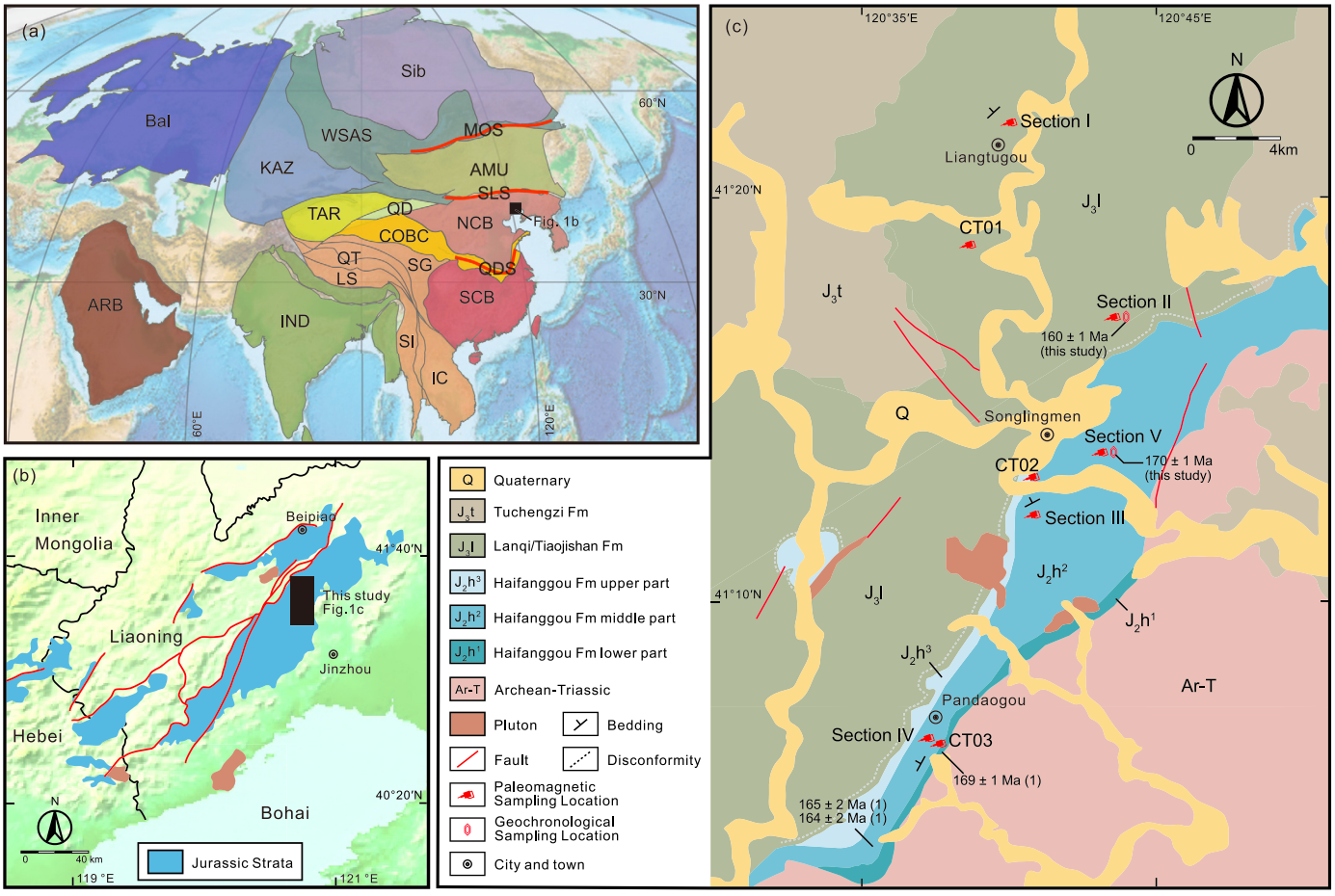


Fig. 1. (a) Tectonic sketch map of Eurasia. The blocks and terrane boundaries are modified from Wang et al. (2004) and Torsvik et al. (2012). (b) Simplified administrative map and distribution of the Jurassic strata in the western Liaoning Province, northern NCB. (c) Simplified geological map of the sampling region. Modified from (BGMRLP, 1989) and Gao et al. (2019). AMU = Amuria block; ARB = Arabia block; Bal = Baltica craton; IC = Indochina block; IND = India craton; KAZ = Kazakhstan block; SIB = Siberia craton; NCB = north China block; SCB = south China block; LS = Lhasa terrane; QT = Qiangtang terrane; SI = Sibumasu block; TAR = Tarim block; WSAS = West Siberia and Altai Sayan; COBC = central orogenic belt of China; SG = Songpan–Ganzi belt; MOS = Mongol–Okhotsk suture; QDS = Qinling–Dabie suture. References for age data: (1) Gao et al. (2019). (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

sic Haifanggou and Lanqi formations are extensively exposed in the region (Figs. 1b and c). The Haifanggou Fm unconformably overlies the Lower Jurassic Beipiao Fm or Middle Triassic sedimentary strata, and can be divided into three rock units: (1) conglomerates and volcanic rocks in the lower part; (2) pyroclastic and volcanic rocks in the middle part; and (3) conglomerates, sandstones and mudstones in the upper part (Figs. 1c and 2). Recently, several U–Pb and $^{40}Ar/^{39}Ar$ ages of Haifanggou Fm in the Beipiao basin have been reported (Fig. 2). An andesite sample in lower part of the Haifanggou Fm was dated at 169 ± 1 Ma (zircon laser ablation-inductively coupled plasma-mass spectrometry [LA–ICP–MS]; Gao et al., 2019). Two U–Pb zircon dates (164 ± 2 Ma and 165 ± 2 Ma) for andesite in the middle part of the Haifanggou Fm were also reported by Gao et al. (2019). And two tuff samples in the middle part of the formation were dated at 167 ± 1 Ma ($^{40}Ar/^{39}Ar$; Chang et al., 2013) and 164 ± 3 Ma (zircon U–Pb LA–ICP–MS; Hao et al., 2020). The Upper Jurassic Lanqi Fm (also called the Tiaojishan Fm in the Luanping region, northern Hebei Province; e.g. Gao et al., 2019; Ren et al., 2016; Yi et al., 2019) unconformably overlies the Haifanggou Fm and is conformably overlain by the Tuchengzi Fm (Fig. 2). The Lanqi/Tiaojishan Fm contains volcanic rocks, intercalated pyroclastics and conglomerate-sandstone sequences. The tuff beds near the bottom of the Lanqi/Tiaojishan Fm were dated at 161 ± 1 Ma and 159 ± 1 Ma ($^{40}Ar/^{39}Ar$; Chang et al., 2009) and 160 ± 3 Ma (zircon U–Pb LA–ICP–MS; Hao et al., 2020). The rhyolite in the upper part of the Lanqi/Tiaojishan Fm was dated at 154 ± 5 Ma (zircon U–Pb LA–ICP–MS; Zhang et al., 2008) (Fig. 2).

In this study, 197 paleomagnetic core samples were collected from 27 sites of red/gray andesite and basalt-andesite lava flows in the lower part of the Lanqi/Tiaojishan Fm at sections I and II (Figs. 3a and b); 72 samples from 9 sites of gray to black basalts in the middle part of the Haifanggou Fm at sections III and IV (Figs. 3c and d); and 180 samples from 20 sites of red/gray andesites and basalt-andesites, as well as gray to black basalts, in the lower to middle part of the Haifanggou Fm at Section V (Fig. 3e). Two block samples were collected to pick out zircons for U–Pb dating near the site GL03 at Section II (GL-NL; Fig. 3b) and site XG06 at Section V (XG-NL; Fig. 3e). The bedding attitudes of strata can be reliably measured from the boundaries between different lava flows, tuffs, gritstones and conglomerates (Fig. 3).

At each site, six to twelve oriented core samples spanning several meters in a stratigraphic thickness were collected. A magnetic compass and a sun compass were used for the orientation. The declination difference between these two orienting methods is within 2°, suggesting that there is no significant local magnetic disturbance in the region. To enhance the reliability of paleomagnetic data, conglomerate tests were performed on interlayered volcanic conglomerates at sites CT01 (in the lower–middle Lanqi/Tiaojishan Fm), CT02 (in the top of the middle Haifanggou Fm) and CT03 (in the base of the middle Haifanggou Fm) (Figs. 1c and 2). The cobbles are petrologically identical to the lava flows that

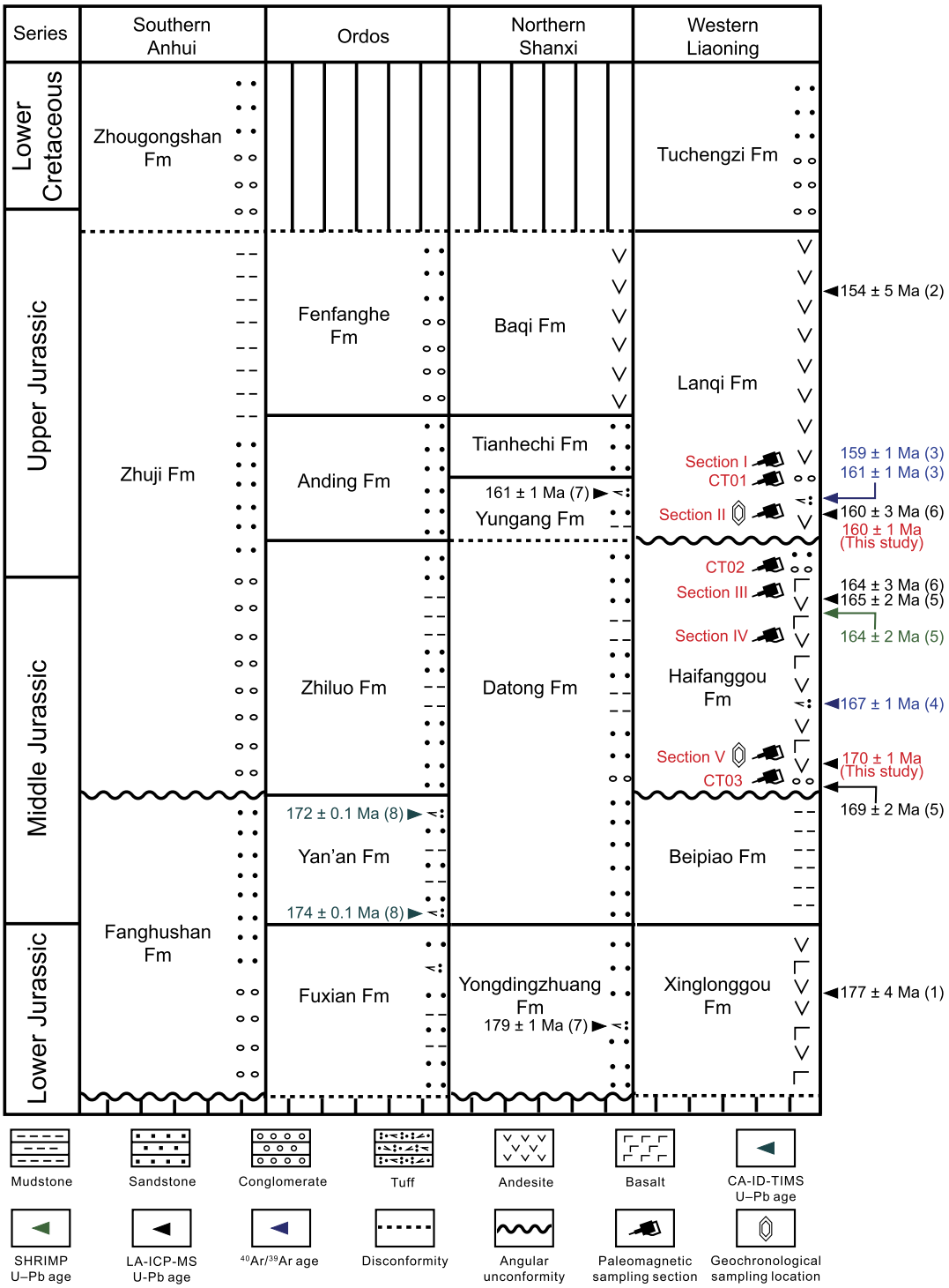


Fig. 2. Simplified late Mesozoic stratigraphic sequences of the NCB, modified from Li et al. (2014); Gao et al. (2019); Hao et al. (2020). Age data from (1) Yang and Li (2008); (2) Zhang et al. (2008); (3) Chang et al. (2009); (4) Chang et al. (2013); (5) Gao et al. (2019); (6) Hao et al. (2020); (7) Li et al. (2014); (8) Zhang et al. (2020). SHRIMP = sensitive high-resolution ion microprobe.

were sampled for paleomagnetic analysis. Ten samples were collected from each site, including eight from different cobbles and two within an individual cobble.

3. Geochronological results

Zircon U-Pb LA–ICP–MS dating was carried out to provide age constraints for the sampling sections (see Appendix A text for method). Zircons extracted from volcanic rocks in the Lanqi/Tiaojishan and Haifanggou formations are mainly euhedral or subhedral and range from 50 to 500 μm in size. Most grains contain the characteristics of idiomorphic crystals such as oscillatory zoning (Figs. 4b and d). Th/U ratios range from 0.6 to 2.7 (Table A.1), indicating the characteristic of magmatic zircons. Twenty-two analyses on the andesite sample GL-NL from the lowest part of the Lanqi/Tiaojishan Fm are concordant and yield a weighted mean $^{206}Pb/^{238}U$ age of 160.4 ± 1.0 Ma (mean square of weighted deviates [MSWD] = 2.9; Fig. 4a). And fourteen analyses on the

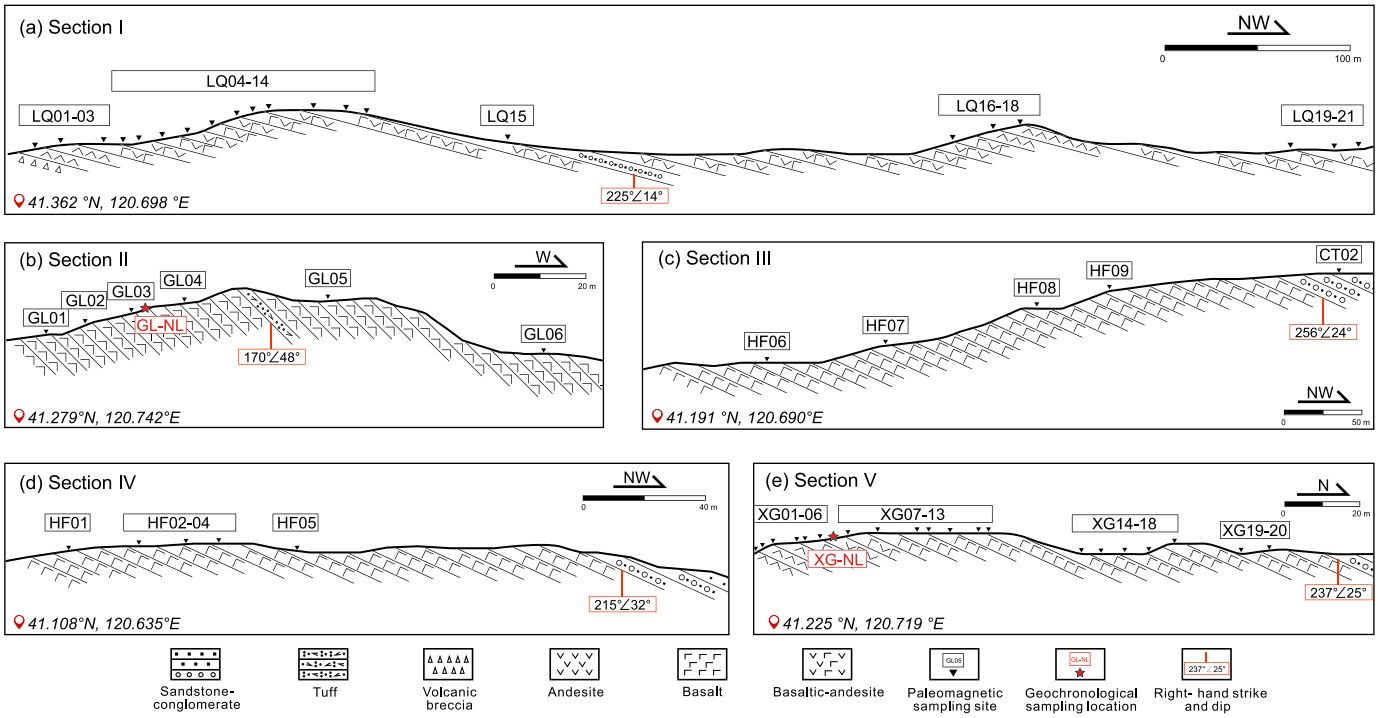


Fig. 3. Measured geological sections showing the locations of the paleomagnetic and geochronological samples.

basaltic-andesite sample XG-NL from the lower–middle Haifanggou Fm are also concordant and give a weighted mean $^{206}Pb/^{238}U$ age of 170.7 ± 1.7 Ma (MSWD = 0.41; Fig. 4c). The new dating results are consistent with the previous age data in the region (Fig. 2; Chang et al., 2009, 2013; Gao et al., 2019; Hao et al., 2020). Therefore, available geochronological results suggest that the age of lower–middle Haifanggou Fm (at Section V) is ca. 170 Ma; the age of the middle Haifanggou Fm (at sections III and IV) is ca. 165 Ma; and the lowest part of the Lanqi/Tiaojishan Fm (at sections I and II) is ca. 160 Ma in age.

4. Paleomagnetism

4.1. Paleomagnetic methods and experimental conditions

Systematic rock magnetic and paleomagnetic experiments and analyses were performed in the Paleomagnetic and Environmental Magnetism Laboratory at China University of Geosciences, Beijing. Each core sample was cut into 1~3 25-mm-long specimens for further measurements. Rock magnetic experiments (see Appendix A text) were applied to representative specimens to identify magnetic carriers and design the demagnetization strategy. All specimens were subjected to stepwise thermal demagnetization by using an ASC TD-48 furnace with in-furnace residual field <10nT. Thermal demagnetization temperature intervals were 50~100 °C for the lower temperature steps and 5~25 °C for the higher temperature steps. Remnant magnetizations were measured through an AGICO JR-6A spinner and a 2G Enterprises 755-4K cryogenic magnetometer. All experiments were conducted in a shielded room with residual magnetic field less than 200 nT. The principal component analysis was utilized to determine the magnetic components (Kirschvink, 1980), and then Fisher statistics was utilized to calculate the site-mean and overall-mean directions (Fisher, 1953). Paleomagnetic data were processed through the programs developed by Enkin (1990) and Cogné (2003), and paleogeographic reconstructions were generated by using the GPlates software (www.gplates.org).

4.2. Paleomagnetic results

All specimens were subjected to stepwise thermal demagnetization, and two magnetic components were identified (Figs. 5a–f). A low temperature component (LTC) was identified below 400 °C (Figs. 5a–f). The *in situ* directions of LTC cluster around the local recent geomagnetic field (RGF) direction in the region (D = 0°, I = 60.3°–61.2°, computed based on the geocentric axial dipole model, Fig. A.2), representing a viscous remnant magnetization (VRM) of the RGF. After removal of the LTC, a high temperature component (HTC) was isolated decaying to the origin between ~400 °C and the unblocking temperature (up to ~580 °C or ~680 °C), which is consistent with the rock magnetic analysis (Appendix A text and Fig. A.1). The HTC directions are defined as the characteristic remanent magnetization (ChRM) directions.

At sections I and II, the ChRM of specimens from the lower part of the Lanqi/Tiaojishan Fm (ca. 160 Ma) was directed SSE with moderate negative inclinations (Figs. 5a, b and g). The site-mean direction for the 27 sites is D = 207.2°, I = −53.8°, k = 17.9, α_{95} = 6.8° *in situ*, and D = 182.6°, I = −53.5°, k = 113.4, α_{95} = 2.6° after tilt correction (Fig. 5g; Table A.2). The ChRMs passed a fold test of McFadden (1990) at 99% confidence level. The optimal concentration of stepwise unfolding for site-level ChRM directions is 111.8 + 5.4/−5.3% (Watson and Enkin, 1993; Fig. 5j; Table A.2), which overlaps with the critical error interval (90%–110%; Meert et al., 2020). The site-mean direction at 111% unfolding (D = 182.6°, I = −53.5°, k = 113.37, α_{95} = 2.6°) is indistinguishable from that at 100% level (Table A.3; Watson and Enkin, 1993), with a small angular separation of 1.4° ± 2.6° at 95% confidence level. These statistical results, together with the latest Jurassic to early Cretaceous folding age (e.g., Hao et al., 2020), lead us to argue that the fold tests satisfy the fold test standard (Meert et al., 2020) and the ChRM represents a pre-folding magnetization. The mean pole 82.7°N/282.6°E (A_{95} = 3.2°) for the lower part of the Lanqi/Tiaojishan Fm was obtained by averaging the virtual geomagnetic poles (VGPs) calculated from the ChRM of each site (Table A.2).

At sections III and IV, the ChRM of specimens from the middle part of the Haifanggou Fm (ca. 165 Ma) was directed NE with

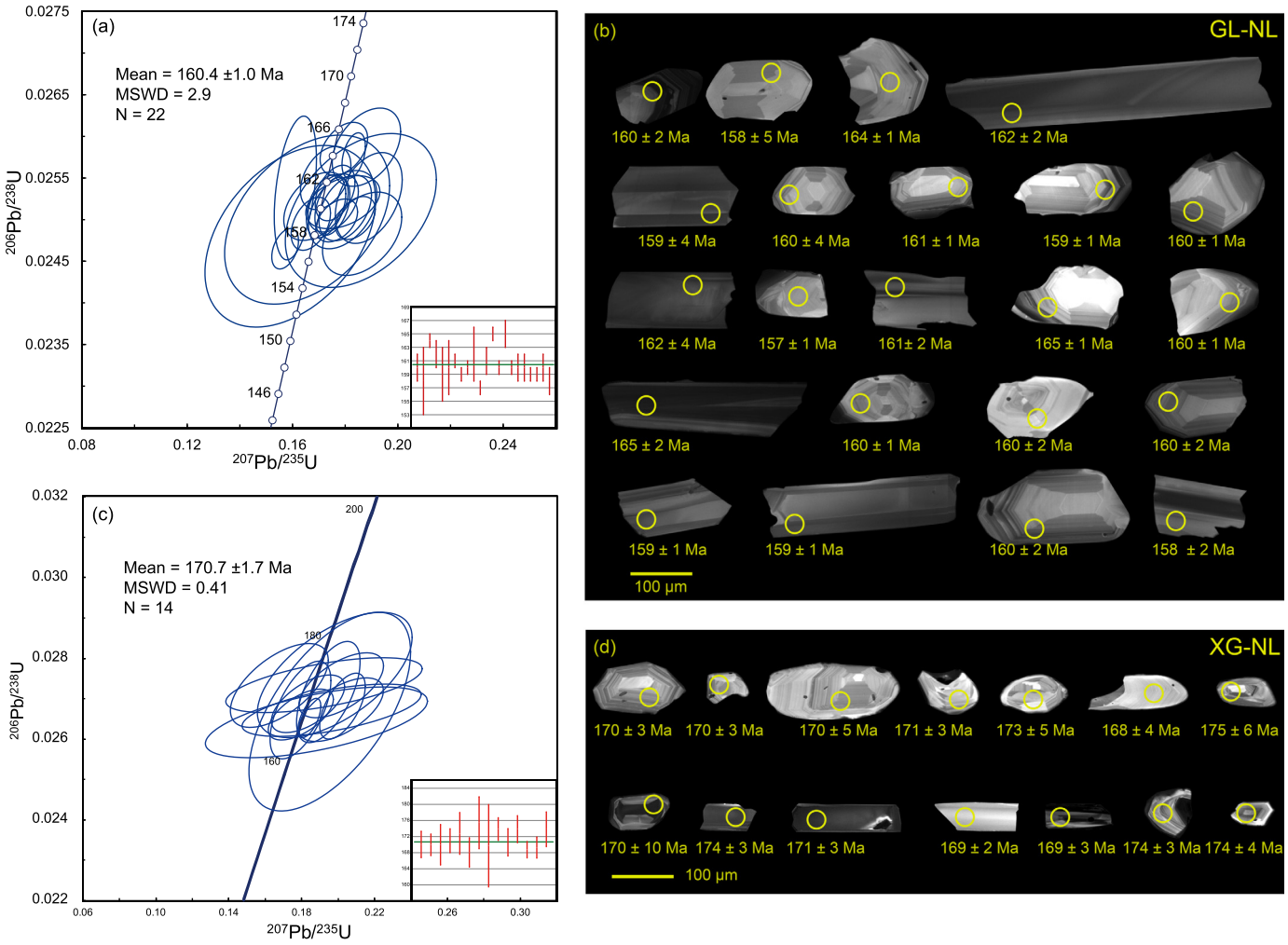


Fig. 4. Cathodoluminescence images of zircons and U-Pb dating results. (a and b) sample GL-NL from the lowest part of the Lanqi/Tiaojishan Fm at the Section II; (c and d) sample XG-NL from the lower–middle Haifanggou Fm at the Section V.

moderate positive inclinations (Figs. 5c, d and h). The site-mean direction for the 9 sites is D = 59.3°, I = 66.6°, k = 82.4, α_{95} = 5.7° *in situ*, and D = 7.5°, I = 58.0°, k = 100.2, α_{95} = 5.2° after tilt correction (Fig. 5h; Table A.2). The fold tests are inconclusive (McElhinny, 1964; McFadden, 1990) as the bedding attitudes between sections change slightly (Fig. 3). The site-mean pole for the 9 sites located at 84.1°N, 228.9°E with A_{95} = 6.6° (Table A.2).

At the Section V, the site-mean ChRM direction for the 20 sites from the lower to middle part of the Haifanggou Fm (ca. 170 Ma) is D = 55.7°, I = 72.9°, k = 49.4, α_{95} = 4.7° *in situ*, and D = 0.9°, I = 60.2°, k = 49.4, and α_{95} = 4.7° after tilt correction (Fig. 5i; Table A.2), corresponding to a paleopole at 89.3°N, 216.8°E, dp = 5.4° and dm = 7.1° (Table A.2).

4.3. Conglomerate tests

Two remanence components were also identified from specimens for conglomerate tests (Figs. 6a and d). The LTC *in situ* directions cluster near the regional RGF direction, representing a VRM of the RGF (Fig. 6b). The specimens from a single cobble show indistinguishable ChRM directions (specimens C-1 and C-2 in Fig. 6c; specimens H-1 and H-2 in Fig. 6e; and specimens G-1 and G-2 in Fig. 6f), indicating the stability of individual cobbles in these conglomerate layers.

At site CT01 of the lower part of Lanqi/Tiaojishan Fm (Fig. 2), the ChRM directions from different cobbles are visually randomly oriented (Fig. 6c). A random distribution test gives R = 2.51 (N = 9), which is significantly less than the critical R-value (R_{crit}) of 4.76 at 95% confidence level (Watson, 1956). At sites CT02 and CT03 of the Haifanggou Fm (Fig. 2), the ChRM directions from different cobbles are also randomly oriented (Figs. 6e and f). The random distribution test gives R = 2.99 (N = 9) for CT02 and R = 2.52 (N = 9) for CT03, which are also less than the R_{crit} of 4.76 (N = 9) at the 95% confidence level (Watson, 1956). These results constitute positive conglomerate tests for the remanence of the Lanqi/Tiaojishan and Haifanggou formations.

4.4. Paleosecular variation analyses

The paleomagnetic records of adjacent volcanic flows showing similar remanence directions may result from the same "spot reading" of geomagnetic field, and large deviation of remanence directions can result from the paleosecular variation (PSV) of the geomagnetic field at any time and location (Biggin et al., 2008; Deenen et al., 2011). Thus, the PSV must be averaged out when a volcanic paleopole is used for tectonic reconstructions. The scatter (S = 8.6° ± 3.3°) of the ca. 160 Ma lower Lanqi/Tiaojishan Fm pole obtained from site VGPs overlaps with the S envelope (S_{max} = 24.7°, S_{min} = 10.3°, Plat = 34.2°; Biggin et al., 2008; Table A.2); and the A_{95} = 1.9° obtained from specimen VGPs is within the A_{95} envelope (A_{95max} = 3.0°, A_{95min} = 1.5°, n = 188; Deenen et al., 2014, 2011; Table A.2). As for the ca. 165 Ma mid-

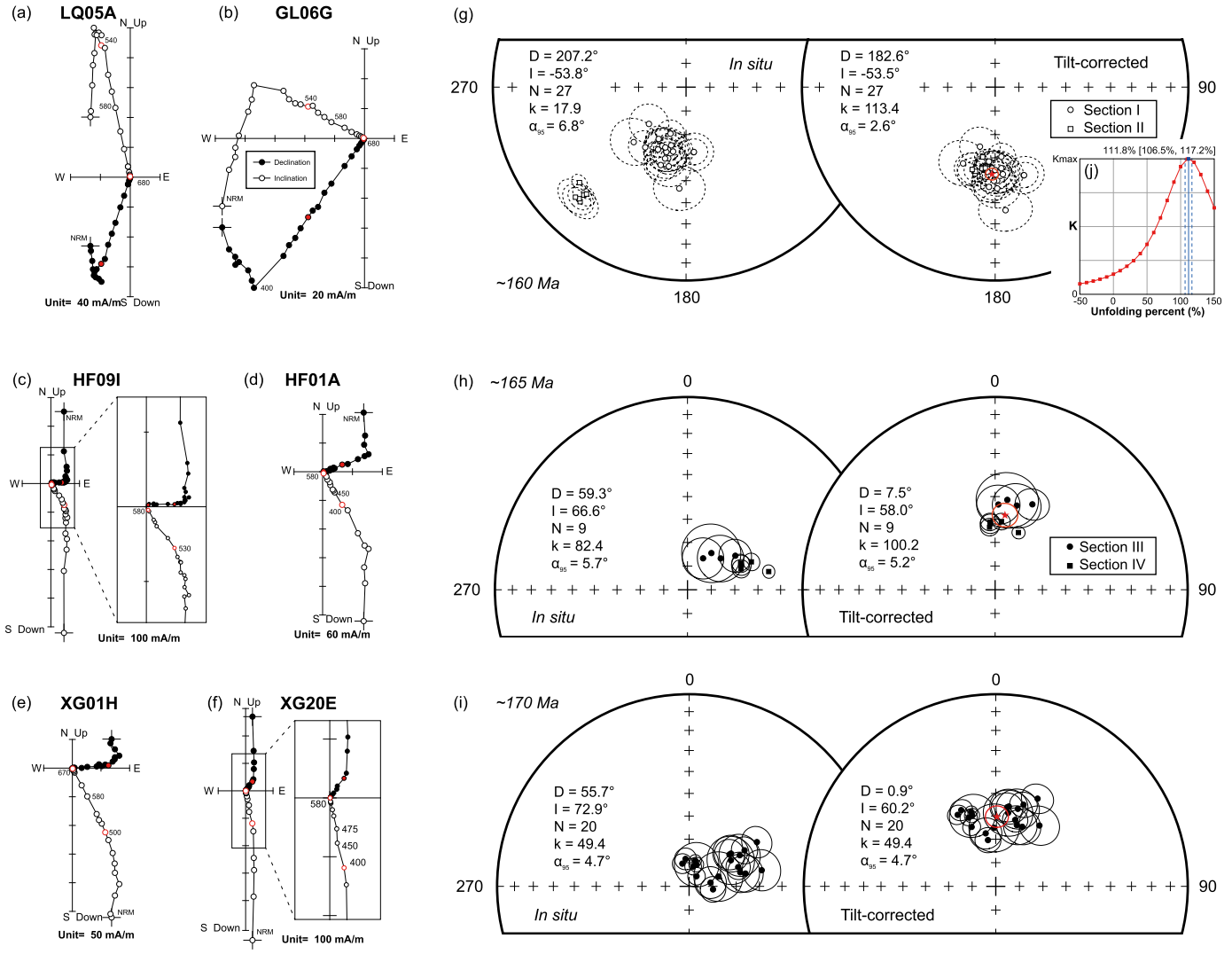


Fig. 5. (a–f) Orthogonal projection plot of representative specimens in geographic coordinates: (a) Section I; (b) Section II; (c) Section III; (d) Section IV; (e) and (f) Section V. Red open/filled circles show demagnetization envelope used in least squares fits of the ChRM (Kirschvink, 1980). NRM = natural remnant magnetization. (g–i) Equal-area projections of the site-mean directions of the ChRM. D = declination; I = inclination; α_{95} = radius of 95% confidence cone of mean direction; k = precision parameter of Fisher (1953); N = number of sites. Filled and open symbols represent downward and upward inclinations, respectively. Stars with 95% confidence circles indicate Fisher statistical direction. (j) Stepwise unfolding analysis (Watson and Enkin, 1993) of the site-level ChRMs for the sections I and II.

dle Haifanggou Fm pole, the scatter S obtained from site VGPs is 10.0° ± 2.2°, which overlaps with the envelope (26.0°, 10.3°, Plat = 38.7°). The A_{95} obtained from specimen VGPs is 3.1°, which is within the envelope of (2.2°, 5.6°, n = 69) (Table A.2). And the scatter (S = 16.1° ± 1.9°) of the ca. 170 Ma lower–middle Haifanggou Fm pole obtained from site VGPs overlaps with the expected value (26.8°, 10.3°) for paleolatitude 41.1°. The A_{95} of this pole obtained from 172 specimen is 2.7°, which is also within the A_{95} envelope (1.5°, 3.2°) (Table A.2). These results suggest the PSV of all the three poles could have averaged out.

In summary, three paleopoles newly obtained from the Jurassic strata in the NCB meet all the seven criteria from the revised seven-point criterion system (Meert et al., 2020): R1, well-dated strata; R2, proper techniques and statistical analyses including positive PSV test; R3, evaluation of the remanence carriers; R4, positive conglomerate tests for the three poles and a positive untilting test for the ca. 160 Ma pole; R5, good structural control; R6, presence of magnetic reversals; and R7, no suspicion of remagnetization based on coincidence with younger poles. Thus, we suggest the paleopoles at 82.7°N, 282.6°E, A_{95} = 3.2° (ca. 160 Ma); 84.1°N, 228.9°E, A_{95} = 6.6° (ca. 165 Ma); and 89.3°N, 216.8°E, dp = 5.4°, dm = 7.1° (ca. 170 Ma) being key poles for the NCB.

5. Discussion

5.1. The NCB had not much paleolatitudinal change during Jurassic

Paleomagnetic results from both continental and oceanic plates suggest that there was a Jurassic TPW event (e.g. Fu and Kent, 2018; Kent et al., 2015; Kent and Irving, 2010; Torsvik et al., 2012), but the timing and duration of the TPW are still controversial (Besse and Courtillot, 2002; Kent and Irving, 2010; Torsvik et al., 2012; Yi et al., 2019). Based on different filtering methods applied to paleomagnetic data, two global apparent polar wander (GAPW) paths have been proposed, which provide alternative views of the Jurassic TPW. Torsvik et al. (2012) used extensive paleomagnetic datasets with quality factors Q ≥3 (Van der Voo, 1990), which included volcanic and sedimentary poles after inclination shallowing corrections. They then computed a GAPW path after applying a quality-weighted mean to the data. This composite path shows a ~30.5° TPW episode that lasts for the entire Jurassic (200–140 Ma)

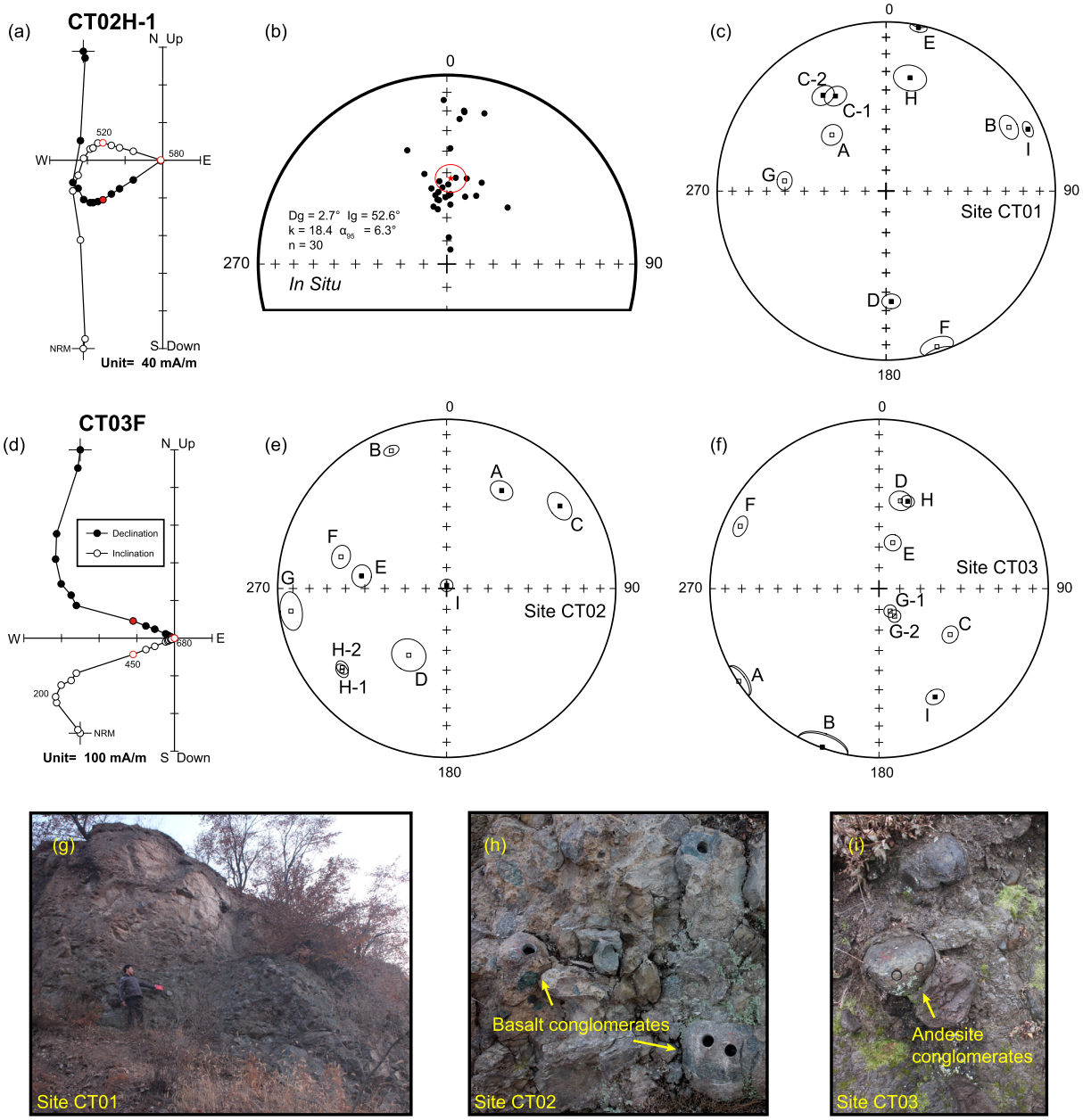


Fig. 6. Conglomerate test results and corresponding photographs of sampling outcrops. (a and d) Orthogonal projection diagrams for representative specimens in geographic coordinates. (b) Equal-area projection of the LTC directions. (c, e and f) Equal-area projections of the ChRM directions of sites CT01, CT02 and CT03, respectively. Specimen ChRM directions are shown by squares with surrounding maximum angular deviation circles. Filled and open squares represent downward and upward inclinations, respectively. (g, h and i) Photographs of sampling outcrops.

and is characterized by relatively lower velocity lithospheric rotation (~0.45–0.8°/myr) around an Euler pole at 0°N, 11°E. These authors, therefore, regarded the Jurassic TPW as a long-term event. Alternatively, Kent and Irving (2010) applied a more stringent quality filter to the database and derived a GAPW path mainly based on high-quality volcanic poles that have precise age constraints. Their results suggest that the Jurassic TPW occurred during 160–145 Ma, and the rotation axis and lithospheric rotation amplitude (~28.6°) were similar to those determined by Torsvik et al. (2012). Due to the shorter event duration, the calculated rotation velocity in Kent and Irving (2010)'s model is as fast as ~2°/myr, which is why the TPW is sometimes referred to as the "monster shift" TPW hypothesis. However, different filtering methods can lead to differential defects. For example, the GAPW path proposed by Torsvik et al. (2012) may average out rapid Earth motion within a limited time interval because of the emphasis on paleomagnetic data from sedimentary rocks. Kent and Irving (2010)'s GAPW path contains fewer poles because of the rigorous screening process that eliminated most of the existing poles. This may lead to a relatively poor record continuity.

The paleomagnetic poles that had been used to establish the GAPW paths all came from the Pangea continent. Eastern Asian blocks, however, were isolated in the Tethys Ocean during Jurassic and did not accrete onto the Eurasia continent until Early Cretaceous (e.g. Huang et al., 2018; Ren et al., 2018; Van der Voo et al., 2015). The paleogeographic location of East Asia was approximately orthogonal to the rotation axis of the Jurassic TPW (Torsvik et al., 2012; Van der Voo et al., 2015), suggesting that the dominant influence of TPW in East Asia was in latitudinal displacement. However, the apparent polar wander (APW) path of

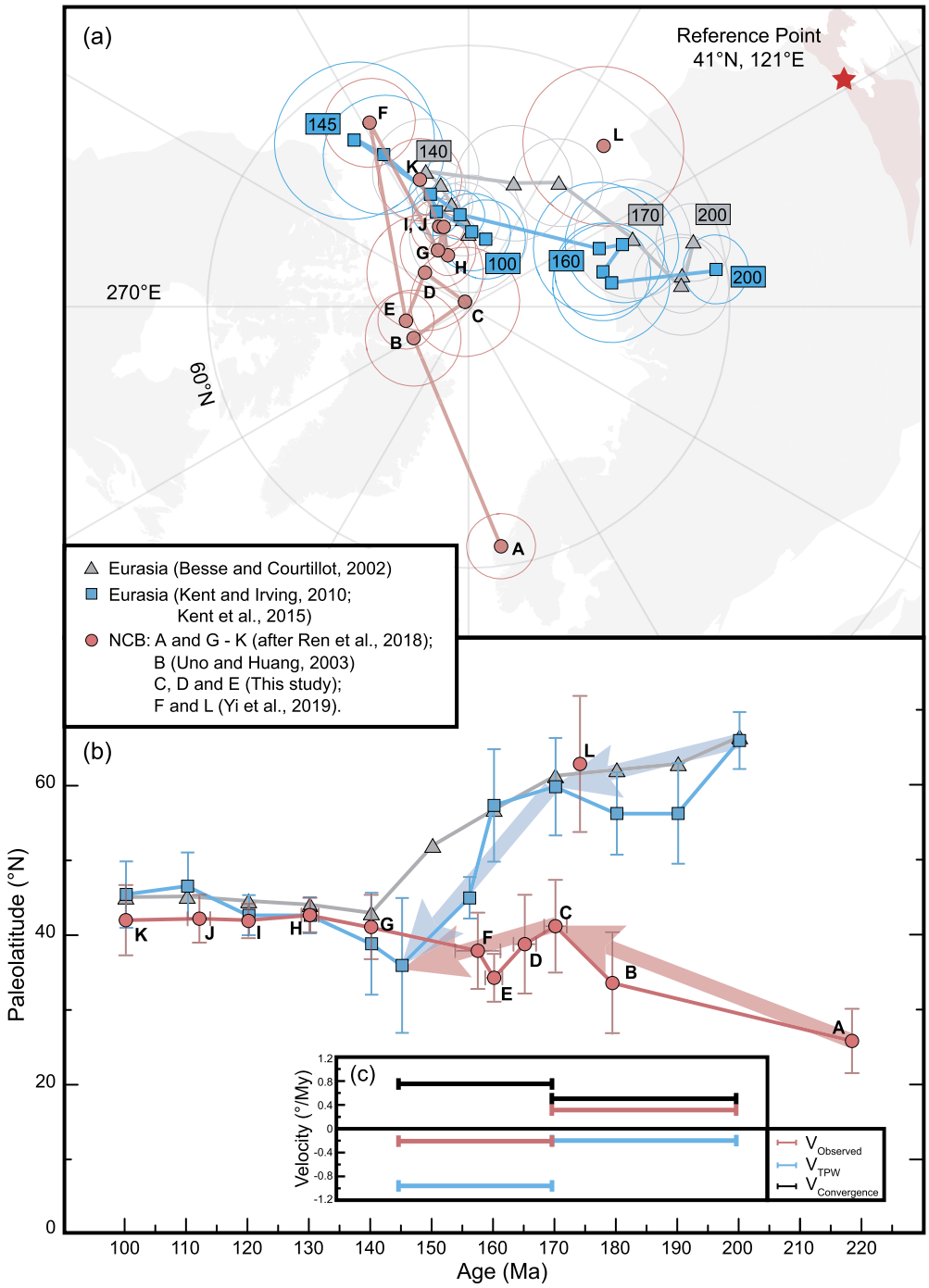


Fig. 7. (a) Mesozoic APW paths for Eurasia and NCB. Poles are listed in Tables 1 and A.3. (b) Paleolatitudinal variations for the Eurasia and NCB calculated by paleopoles from their APW paths at the reference point (41°N, 121°E) in the NCB. (c) Calculated velocity of TPW (V_{TPW}), observed motion velocity of NCB ($V_{Observed}$) and net northward plate movement velocity of NCB after removing TPW component ($V_{convergence}$). Positive for northward.

the eastern Asian blocks, including the AMU-NCB, does not record the same large-scale (>25°) latitudinal motions as required by the TPW hypotheses, regardless of whether models based on extensive datasets or selected key poles are considered (e.g. Huang et al., 2018; Ren et al., 2018; Van der Voo et al., 2015).

Yi et al. (2019) proposed that the NCB, part of East Asia, was located at a high paleolatitudinal region (~62.8°N for the reference point 41°N, 121°E) at ca. 174 Ma based on the paleomagnetic pole newly obtained from the Nandaling Fm (ca. 180–168 Ma; Yi et al., 2019 and references therein), and suggested that the NCB had experienced a "monster shift" TPW between 174–157 Ma. However, the Nandaling pole is abnormal to other coeval paleomagnetic poles published previously and from this study (Fig. 7). A Lower Jurassic pole obtained from the sedimentary rocks of the Fuxian Fm in the Ordos basin and the correlative Yongdingzhuang Fm in the northern Shanxi region in central NCB (Table 1; Fig. 7; Uno and Huang, 2003) can be constrained around ca. 179 Ma on the basis of the age of zircons in pyroclastic rocks (179 ± 1 Ma) in the middle Yongdingzhuang Fm (Fig. 2; Li et al., 2014). This pole indicates the paleolatitude of the reference point (41°N, 121°E) is ~32.5°N (Fig. 7b). If we perform an inclination shallowing correction with an assumed flattening factor f = 0.6 for the pole, the

Table 1
Selected Late Triassic to Cretaceous poles from the north China block.

Age (Ma)	Type	N(S)	Plat (N°)	Plon (E°)	dp/dm A_{95} (°)	R	References
K_{1-2} (ca. 100)^{K}	S	10	74.5	201.0	4.7	5	Gilder and Courtillot (1997)
114–110^{J}	L	31	80.3	200.3	3.2	6	Zhu et al. (2008)
120 mean^{I}	L	(5)	83.7	202.0	5.1	-	Ren et al. (2018)
130^{H}	L+S	35	80.5	197.4	2.3	7	Ren et al. (2018)
140^{G}	S	21	82.7	208.6	4.3	7	Ren et al. (2018)
157^{F}	L	51	66.1	208.3	5.1	7	Yi et al. (2019)
160^{E}	L	27	82.7	282.6	3.2	7	This study
165^{D}	L	9	84.1	228.9	6.6	7	This study
170^{C}	L	20	89.3	216.8	5.4/7.1	7	This study
180–168^{L}	L+S	24	65.9	140.3	9.1	7	Yi et al. (2019)
179^{B}	S	15	81.5	302.5	5.5	6	Uno and Huang (2003)
T_3 (ca. 219)^{A}	S	11	62.3	7.7	3.8	5	Yang et al. (1998)

Note. K_{1-2} = Early-Late Cretaceous; T_3 = Late Triassic; $^{A-L}$ = codes for poles, same as Fig. 7; see Appendix A text for more details; Type = type of the lithology, S = sedimentary, L = lava; N(S) = number of sites (studies); Plat = latitude of poles; Plon = longitude of poles; A_{95} = 95% confidence limit of poles; R = reliable criterion factor (R-Score; Meert et al., 2020).

paleolatitude becomes ~46.1°N, which is still obviously lower than the high paleolatitudinal results of the Nandaling Fm. Recently, the Middle Jurassic Yan'an Fm, conformably overlying the Fuxian Fm, was dated at 173.9–171.7 Ma using the zircon U-Pb chemical abrasion-isotope dilution-thermal ionization mass spectrometry (CA–ID–TIMS) method on interstratified ash beds (Fig. 2; Zhang et al., 2020). The new geochronologic data provide a precise age constrain for the pole of the Yan'an Fm (Yang et al., 1998). Based on the Yan'an pole, the paleolatitudes without and after inclination correction with f = 0.6 of the reference point are calculated to be ~33.9°N and ~47.1°N, respectively. Both of them are also lower than the paleolatitude calculated based on the Nandaling pole. Furthermore, the Nandaling pole is also inconsistent with the ca. 170 Ma and ca. 165 Ma poles obtained in this study (Fig. 7b). Accordingly, we suggest that the Nandaling pole cannot be used to calculate the paleolatitude of the NCB around 174 Ma and the single anomalous paleomagnetic pole should be treated with caution for tectonic interpretations.

Although quality of the Nandaling pole remains to be tested, the new findings highly impacted the paleomagnetic database of the NCB. Their new model has greatly challenged our knowledge of the Mesozoic paleogeography of the eastern Asian blocks and therefore has evoked great interest of further test. In this study, our new high-quality paleomagnetic and geochronological results from the ca. 170–160 Ma volcanic rocks, together with the ca. 157 Ma pole of the Tiaojishan Fm (Yi et al., 2019), indicate that the paleolatitude of the sampling region in the NCB was between 34.2–41.1°N during the Jurassic TPW time interval (Fig. 7b) and suggest that the NCB had not experienced any large-scale latitudinal geographic change between ca. 170–157 Ma. Such a paleogeographic "monster shift" proposed in Yi et al. (2019) was also unlikely pre-170 Ma in Jurassic, if takes all high-quality paleomagnetic data into account.

5.2. Convergence of the MOO in the Jurassic TPW framework

Withstanding the odd paleomagnetic data of the Nandaling Fm, we would rather offer a model more like the traditional ones. In the following discussion, we adopt the revised GAPW path from Kent and Irving (2010) to represent the motion of TPW in the Eurasian coordinate (Kent et al., 2015), and plot an alternative GAPW path derived from a similar database to that used by Torsvik et al. (2012) without the inclination error correction (Besse and Courtillot, 2002) (Fig. 7a; Table A.2). We also refined the APW path for the NCB through the use of the updated database and the revised quality criteria (Meert et al., 2020) (see Appendix A text, Fig. 7a and Table 1), which has a similar selection strategy to that used by Kent and Irving (2010).

During the Jurassic TPW event, Eurasia and eastern Asian blocks underwent a large-scale convergence, which involved shortening and subduction of the MOO (Ren et al., 2016, 2018; Van der Voo et al., 2015). Paleogeographic movements of the eastern Asian blocks in this critical time interval should thus record both the TPW and plate motion components. Some attempts have been made to determine the "absolute plate motion" of Eurasia and the AMU-NCB during convergence (Van der Voo et al., 2015; Wu et al., 2017a). However, prior to this study, successive and robust Middle to Late Jurassic (170–160 Ma) paleomagnetic data have not been available to identify the TPW component for the eastern Asian blocks.

To analyze the relative kinematic relationships of Eurasia and AMU-NCB, we translated the APW paths into paleolatitude variation and motion velocity graphs (Figs. 7b and c) at a reference point (41°N, 121°E) in the NCB. Our model suggests that latitudinal convergence between Eurasia and the NCB initiated in the Late Triassic and ceased no later than 140 Ma, consistent with previous studies (e.g. Ren et al., 2018; Van der Voo et al., 2015). From 200 to 170 Ma, both paleolatitudinal variation curves of Eurasia show similar initial and final latitudes at 65.9°N and 59.8°N, respectively (Fig. 7b), although the model of Kent et al. (2015) shows additional fluctuations compared to Besse and Courtillot (2002). This southward motion of Eurasia may suggest the Jurassic TPW have already initiated at ca. 200 Ma, which was interpreted as the beginning of Jurassic long-term TPW by Torsvik et al. (2012). During this period, the velocity of Eurasia is approximately −0.2°/myr (negative for southward), also representing the velocity of TPW (V_{TPW}). While the NCB's paleolatitudinal variation shows a continuous increase from 26.2°N to 41.1°N from Late Triassic (ca. 219 Ma) to ca. 170 Ma. The observed velocity of the NCB ($V_{Observed}$) in geological coordinates is ~0.30°/myr. After removing the component of TPW, a net northward plate moving velocity of the NCB ($V_{Convergence}$) is calculated as ~0.50°/myr (Fig. 7c). On this basis, we conclude that there had been ~3800 km of shortening between Eurasia and the NCB since 200 Ma, and that this shortening was reduced to ~2000 km by 170 Ma.

The motion characteristics of the two plates changed after 170 Ma. Southward shift of Eurasia accelerated and is referred to as the "monster shift" TPW by Kent and Irving (2010) (Fig. 7b). While the NCB also recorded southward motions in this time interval (Fig. 7b). The paleolatitude of the two continents became indistinguishable around 140 Ma, indicating that the amalgamation between Eurasia and AMU-NCB had been already achieved at that time (Fig. 7b). However, due to a lack of key poles in the NCB between 157 Ma and 140 Ma, the precise timing of amalgamation

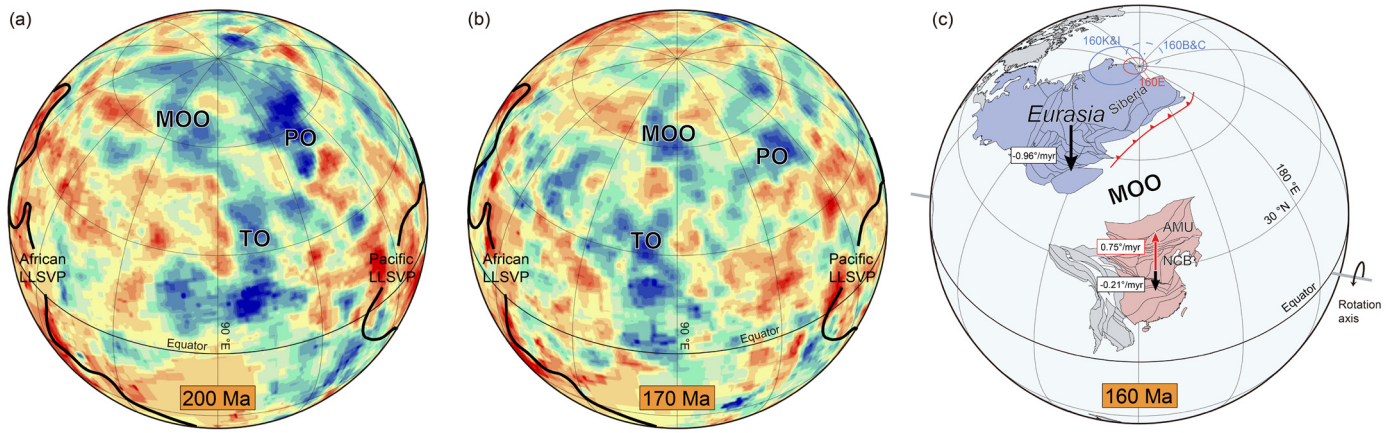


Fig. 8. (a and b) TPW corrected seismic tomography horizontal slices at 2410 km (ca. 200 Ma) and 2050 km (ca. 170 Ma) projected to the Earth surface, respectively (UU-P07 model; Amaru, 2007). The depth of seismic slice was calculated using the ~12 mm/yr average slab sinking rate following van der Meer et al. (2010). TPW correction Euler parameters are listed in Table A.5 (modified from Torsvik et al., 2012). TO = Tethys Ocean; PO = Pacific Ocean. (c) Sketch paleogeographic reconstruction of the Eurasia, Eastern Asian blocks and other main plates in the Jurassic TPW coordinate (Torsvik et al., 2012) at 160 Ma. "160E" represents the ca. 160 Ma pole of NCB obtained in this study (Table 1). "160K&I" and "160B&C" represent 160 Ma poles of Eurasia in Kent and Irving (2010) and Besse and Courtillot (2002), respectively (Table A.4). Configurations of main blocks in Pangea and eastern Asia are following Torsvik et al. (2012) and Huang et al. (2018), respectively. For Euler rotation parameters, see Table A.6. The observed/net latitudinal movement velocities of plates ($V_{Observed}/V_{convergence}$) are represented by black/red arrows (reference point at 41°N, 121°E). See section 5.2 and Fig. 7 for more details.

remains uncertain. Because the track of poles of NCB between 170 Ma and 157 Ma points directly to the 145 Ma Eurasian pole, we suggest that Eurasia and AMU-NCB had merged by 145 Ma, and use the 145 Ma Eurasian pole to calculate the velocity of the NCB. The V_{TPW} is approximately −0.96°/myr, and the $V_{Observed}$ is approximately −0.21°/myr. Therefore, the $V_{Convergence}$ is calculated as ~0.75°/myr (Fig. 7c), and the convergent distance of the NCB during 170–145 Ma is estimated as ~2000 km.

In summary, we recognized an initiation and boosting processes of the Jurassic TPW at the reference point in the NCB, which belonged to different coordinate systems and tectonic domains from the Pangea continent during Jurassic. Based on these observations, we determined a relatively constant net northward plate moving velocity for the NCB in the whole Jurassic after removing the effects of TPW. This dynamic model of the NCB, which is similar to previous geological and tectonic models (Huang et al., 2018; Ren et al., 2018; Van der Voo et al., 2015) and also satisfies the Jurassic TPW hypotheses (Kent and Irving, 2010; Torsvik et al., 2012), suggesting the presence of TPW can be perfectly illustrated in East Asia. The continuous northward motion of NCB in the Jurassic is supported by seismic tomography analysis evidence of the northward-moving MOO slab (Figs. 8a and b; Van der Voo et al., 2015; Wu et al., 2017b), and by the eruption of voluminous subducted-related basaltic lavas along the southern margin of Siberia craton (Fig. 8c; Ren et al., 2018 and references therein).

5.3. Possible geodynamic triggers of the Jurassic TPW

Several assumptions have been proposed to explain the trigger of the Jurassic TPW, including the growth of continental ice sheets, the development of large igneous provinces, and the descent of subducted slabs (e.g. Fu et al., 2020; Fu and Kent, 2018; Kent et al., 2015). Numerical models suggested that persistent subductions with proper sinking rate, even along a single subduction zone, are potentially sufficient to induce the TPW (Fu et al., 2020 and references therein).

Global mantle tomography model has been used to estimate the oceanic slabs subducted in ancient ages (e.g. Amaru, 2007; van der Meer et al., 2010; Van der Voo et al., 2015). However, the locations of the oceans estimated from the tomography of modern Earth may not be the same as their original sinking locations due to the rotation of entire crust-mantle system caused by the TPW. For the Jurassic TPW, although its manifestation pattern is still under debate, there is an agreement about the total rotational amplitude (Fu and Kent, 2018; Kent et al., 2015; Torsvik et al., 2012). We thus revised the Jurassic segment of the finite reconstruction rotation of TPW correction proposed by Torsvik et al. (2012) based on our observation in East Asia and applied it to the seismic tomographic slices (Figs. 8a and b).

The TPW is more likely to be triggered by long-term accumulation of mass anomaly (Evans, 2003). During the late Mesozoic, active oceanic subductions were in a boom phase (Amaru, 2007; van der Meer et al., 2010). After TPW correction, the mantle seismic tomographic slices at ca. 200 Ma and ca. 170 Ma indicate that the existence of two Large Low Shear-wave Velocity Provinces (LLSVPs) locked the TPW axis (minimum inertial axis of the Earth) through Africa and the antipodal Pacific (Figs. 8a and b; Torsvik et al., 2012); while the compact and rigid ancient crusts distributed on the east part of Northern Hemisphere and near the longitudes roughly orthogonal to the TPW axis are interpreted as the MOO, Pacific Ocean, and perhaps Tethys Ocean (Figs. 8a and b; Amaru, 2007; van der Meer et al., 2010; Van der Voo et al., 2015). The oceanic subducted slabs do not sink directly into the lower mantle along the subduction zones, but are often deflected by the 660 km mantle discontinuity and apparently stagnate within the mantle transition zone (e.g. Fukao and Obayashi, 2013). The preservation of detached oceanic slabs in the upper mantle since pre-Jurassic might lead to an increasing of positive mass anomaly in an advantageous position to initiate the Jurassic TPW (Fig. 7). As the oceanic slabs continuously sunk, the Jurassic TPW entered a boosting episode after ca. 170 Ma (Fig. 7). The overloaded oceanic slabs on the eastern side of Earth might be transferred and be released into the deep mantle, and eventually resulted in a slab avalanche event in southeastern Asia in the early Late Cretaceous (ca. 90 Ma; Rasskazov and Chuvashova, 2017). Additionally, the path of TPW showed an oscillation roughly orthogonal to the TPW axis from 200 Ma to present (Doubrovine et al., 2012; Torsvik et al., 2012), which may be a response to the extreme mass of two LLSVPs and the constant variations of active oceans and subducted slabs in favorable positions.

The convergence of the W-E trending MOO may play an irreplaceable role in the inducement of the Jurassic TPW (Fig. 8). The TPW corrected seismic tomographic images indicate that the extension of the MOO slabs may have reached higher than 80°N at

ca. 200 Ma (Fig. 8a; Amaru, 2007; van der Meer et al., 2010; Van der Voo et al., 2015). The regional positive mass anomaly caused by wide MOO subducted slabs in the higher latitudes should have a larger torque than the dispersed distribution in lower latitudes to trigger the Jurassic TPW easier. The subduction of MOO along the southern margin of Siberia have initiated at least in the early Triassic (Kravchinsky et al., 2002; van der Meer et al., 2010; Van der Voo et al., 2015). The convergence of MOO was still ongoing and more than 3800 km oceanic crusts have subducted beneath the Siberia craton in Jurassic, leading to an accumulation of excess mass in high paleolatitudes which contributed to the Jurassic TPW. Especially during 200–170 Ma, the oceans in the lower latitudes mainly showed migrations around the longitudes roughly orthogonal to the TPW axis, however, the oceanic crusts in the high latitudes experienced significant substantial reductions (Figs. 8a and b). Furthermore, the ending of Jurassic TPW was consistent in time with the amalgamation between Eurasia and Eastern Asia in the earliest Cretaceous (Fig. 7), which may also imply a close relationship between them.

Meanwhile, the evolution of oceanic subductions should be accompanied by the upwelling and downwelling mantle flows. We do not deny the contribution of mantle conventions which were in concert with the subductions, and other potential driving forces generated in the same time interval.

6. Conclusions

Three robust poles were obtained from the Middle and Late Jurassic volcanic rocks in the NCB. Quality of them was enforced by positive untilting test, conglomerate test, the presence of magnetic reversals and secular variation analyses. The updated paleomagnetic database suggests that the NCB accreted to Eurasia around 140 Ma and did not undergo any large southward paleogeographic shift in Jurassic. After TPW correction, a net northward plate movement velocity of the NCB was calculated at 0.50–0.75°/myr in Jurassic. The long-term accumulation of MOO subducted slabs at the high paleolatitudes may induce the initiation of Jurassic TPW at ca. 200 Ma and drive it into a boosting episode after ca. 170 Ma.

CRediT authorship contribution statement

Yangjun Gao: Conceptualization, Investigation, Writing – original draft. **Shihong Zhang:** Conceptualization, Investigation, Resources, Writing – original draft. **Hanqing Zhao:** Investigation, Writing – review & editing. **Qiang Ren:** Investigation, Writing – review & editing. **Tianshui Yang:** Resources, Writing – review & editing. **Huaichun Wu:** Resources, Writing – review & editing. **Haiyan Li:** Resources, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. Supplementary material

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